

DAX Functions in Power BI

Power BI Assignment - Complete DAX Implementation

Dataset	amazon_store_sales_2022.csv
Records	500
Date Range	2022-01-01 to 2023-05-15
Power BI Table Name	amazon_store_sales_2022

This assignment uses calculated columns and measures in DAX exactly as they would be created in Power BI Desktop. Results shown below are calculated from the supplied dataset.

Question 1

Create a calculated column **TotalPrice** using Quantity and Unit Price.

Solution

Power BI implementation:

Modeling -> New column

```
TotalPrice =  
'amazon_store_sales_2022'[Quantity] * 'amazon_store_sales_2022'[UnitPrice]
```

This calculated column multiplies Quantity by UnitPrice for every row.

OrderID	Quantity	UnitPrice	TotalPrice
10001	9	495.78	4462.02
10002	8	35.45	283.60
10003	6	442.04	2652.24
10004	7	260.46	1823.22
10005	3	453.72	1361.16

Result: Total of the calculated TotalPrice column = 634,715.73

Question 2

Create a calculated column **DiscountedPrice**.

Solution

Power BI implementation:

Modeling -> New column

```
DiscountedPrice =  
'amazon_store_sales_2022'[TotalPrice] * (1 - 'amazon_store_sales_2022'[Discount])
```

Discount is stored as a decimal (for example 0.20 = 20%). The formula subtracts the discount percentage from TotalPrice.

OrderID	TotalPrice	Discount	DiscountedPrice
10001	4462.02	4%	4283.54
10002	283.60	29%	201.36
10003	2652.24	1%	2625.72
10004	1823.22	22%	1422.11
10005	1361.16	23%	1048.09

Result: Total DiscountedPrice = 540,429.62

Question 3

Create a measure **Average Order Value (AOV)**.

Solution

Power BI implementation:

Modeling -> New measure

```
Average Order Value (AOV) =  
DIVIDE(  
SUM('amazon_store_sales_2022'[SalesAmount]),  
DISTINCTCOUNT('amazon_store_sales_2022'[OrderID]),  
0  
)
```

This measure divides total sales by the number of distinct orders, so it responds dynamically to filters and slicers.

Result: Average Order Value = 2,516.08

Total Sales	Distinct Orders	AOV
1,258,041.61	500	2,516.08

Question 4

Using DAX, create a measure that calculates cumulative Year-to-Date (YTD) sales based on the Order Date.

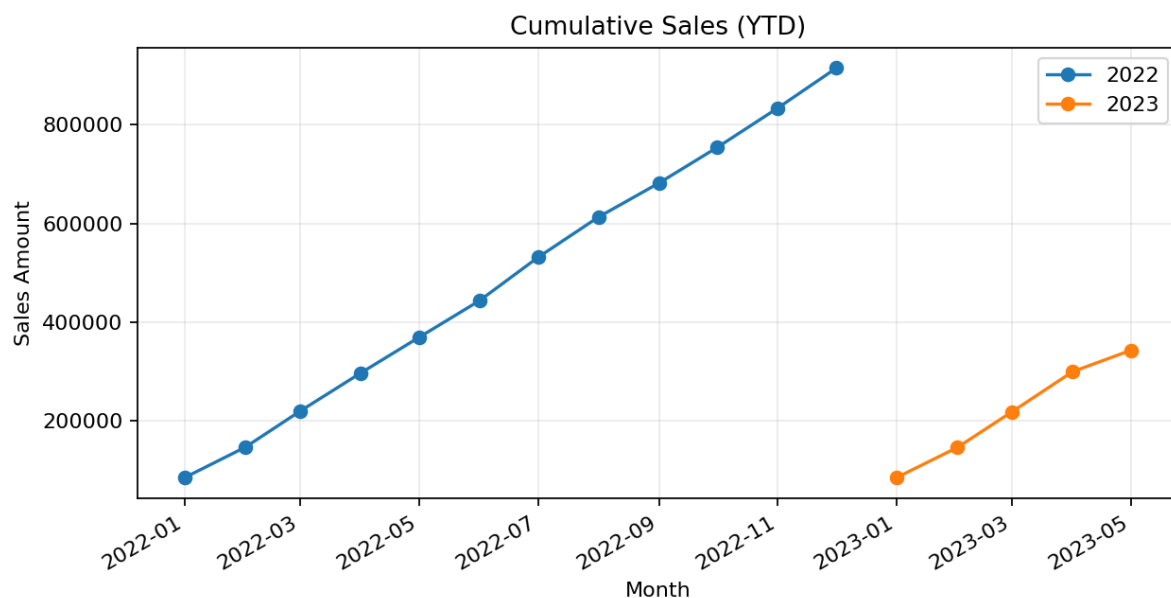
Solution

Power BI implementation:

```
Total Sales =  
SUM('amazon_store_sales_2022'[SalesAmount])  
  
Sales YTD =  
TOTALYTD(  
[Total Sales],  
'amazon_store_sales_2022'[OrderDate]  
)
```

The Sales YTD measure returns cumulative sales from the beginning of the current year up to the date in the current filter context.

Result at latest date (2023-05-15): YTD Sales = 343,142.21



Question 5

Using DAX, create a measure to rank each product according to its Total Sales, with the highest-selling product ranked first.

Solution

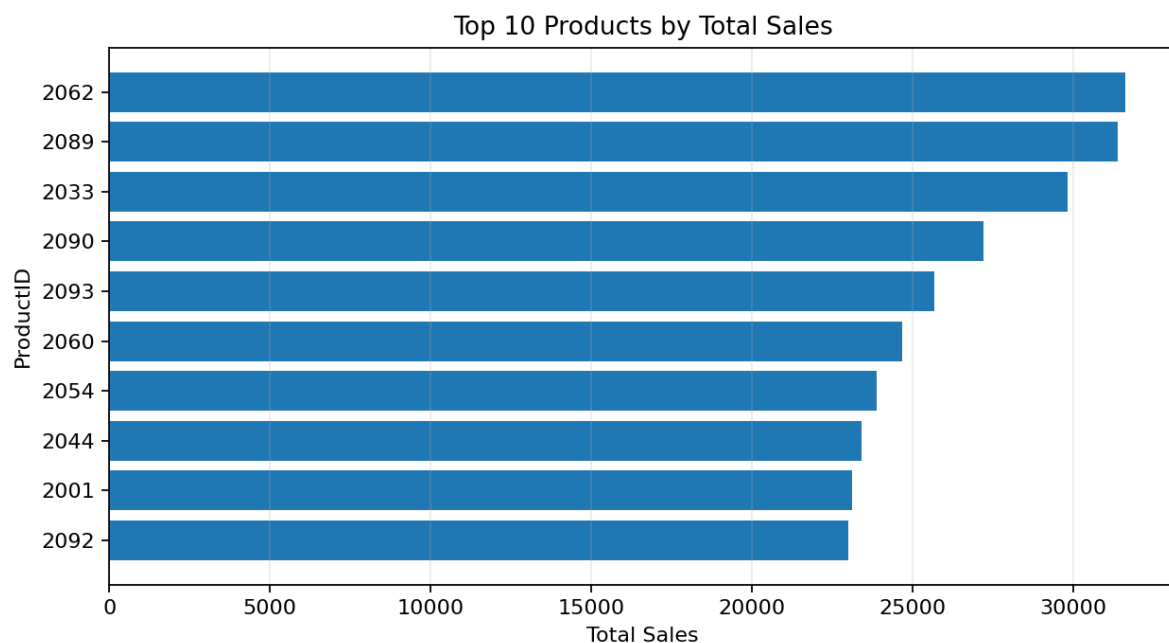
Power BI implementation:

```
Product Rank =  
RANKX(  
ALL('amazon_store_sales_2022'[ProductID]),  
[Total Sales],  
,  
DESC,  
DENSE  
)
```

RANKX evaluates Total Sales for all ProductID values. DESC puts the highest-selling product at rank 1, and DENSE avoids gaps in ranking.

Rank	ProductID	Total Sales
1	2062	31,622.32
2	2089	31,385.84
3	2033	29,834.37
4	2090	27,209.72
5	2093	25,670.84

Result: ProductID 2062 is ranked 1 with Total Sales of 31,622.32.



Question 6

Create a calculated column classifying orders as **High Value** if sales > 2000, else **Low Value**.

Solution

Power BI implementation:

```
Order Value Category =  
IF(  
    'amazon_store_sales_2022'[SalesAmount] > 2000,  
    "High Value",  
    "Low Value"  
)
```

The calculated column evaluates each row independently and labels the order according to SalesAmount.

Category	Number of Orders
High Value	308
Low Value	192

Result: 308 orders are High Value and 192 orders are Low Value.

Question 7

Create a measure to calculate **total sales from the Online channel only**.

Solution

Power BI implementation:

```
Online Sales =  
CALCULATE(  
[Total Sales],  
'amazon_store_sales_2022'[SalesChannel] = "Online"  
)
```

CALCULATE changes the filter context so the measure includes only rows where SalesChannel is Online.

Result: Online Sales = 677,525.24

Sales Channel	Total Sales
Mobile App	580,516.37
Online	677,525.24

Question 8

Create a measure called **Date Difference** that calculates the number of days between the latest Order Date and today's date using the DATEDIFF function.

Solution

Power BI implementation:

```
Date Difference =  
VAR LatestOrderDate = MAX('amazon_store_sales_2022'[OrderDate])  
RETURN  
DATEDIFF(  
    LatestOrderDate,  
    TODAY(),  
    DAY  
)
```

MAX(OrderDate) returns the latest order date in the current filter context. TODAY() supplies the current date, and DATEDIFF returns the difference in days.

Result as of 2026-09-12: Latest Order Date = 2023-05-15 and Date Difference = 1,216 days.

Final DAX Measure Summary

Object	Type	Purpose
TotalPrice	Calculated Column	Quantity x UnitPrice
DiscountedPrice	Calculated Column	Price after discount
Average Order Value (AOV)	Measure	Average sales per order
Sales YTD	Measure	Cumulative sales within year
Product Rank	Measure	Rank products by sales
Order Value Category	Calculated Column	High/Low value classification
Online Sales	Measure	Sales for Online channel only
Date Difference	Measure	Days since latest order